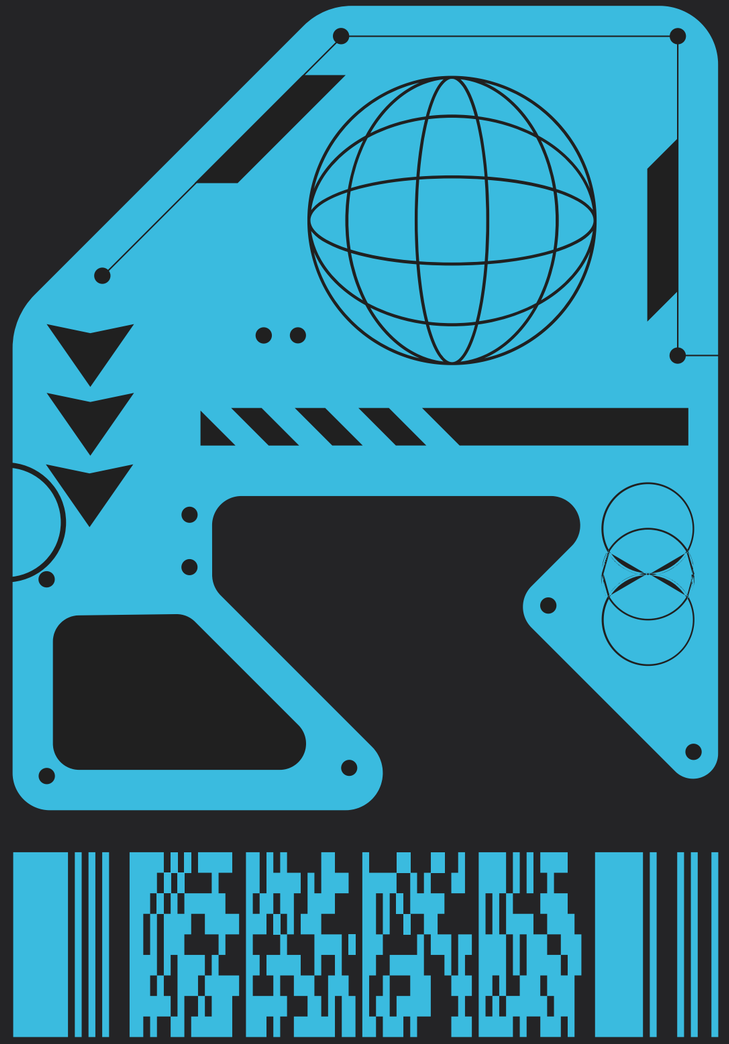
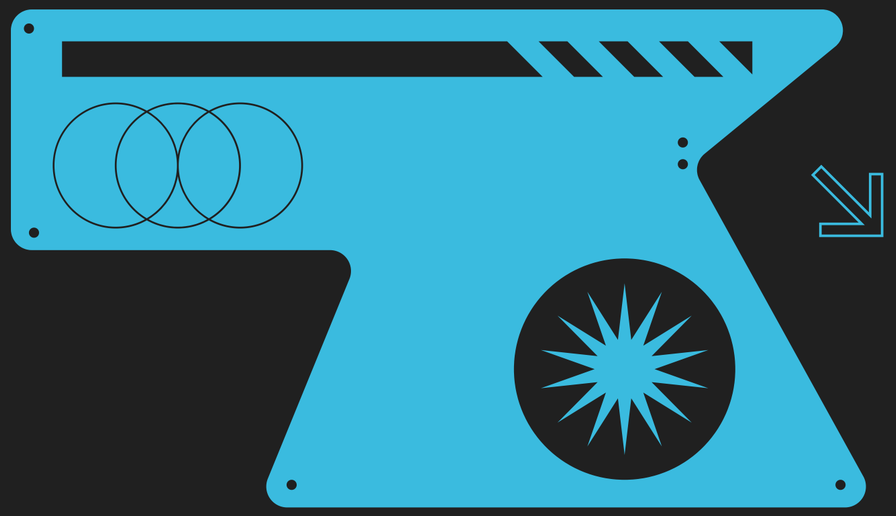
NOT SO PHISHY

BRIEF

An ML based Real-Time URL classifier that identifies benign and malicious URLs based on attributes of the URL after passing them through the model.



TEAM TECH FUSION



MEET THE TEAM

AKSHAY NEGI

E22CSEU0264

Contributed in Model Training and Feature Extraction

ADITI CHATURVEDI

E22CSEU0252

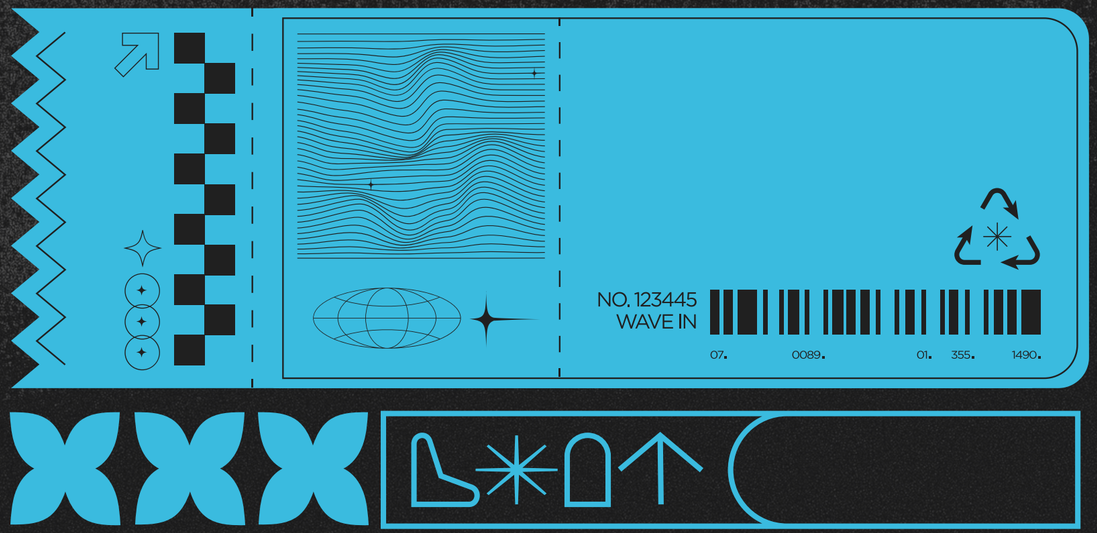
Contributed in Model Training and Backend Integration

SATYAM RAJ

E22CSEU0257

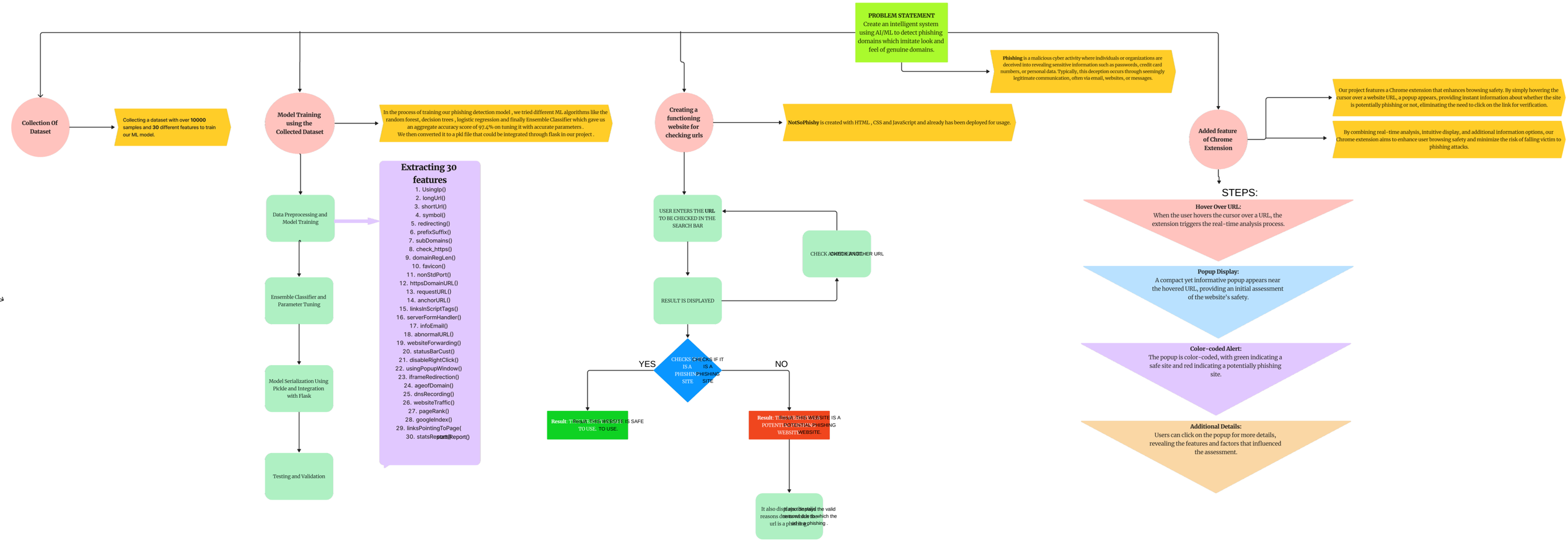
Contributed in Model Training and Frontend Development

PROJECT DESCRIPTION



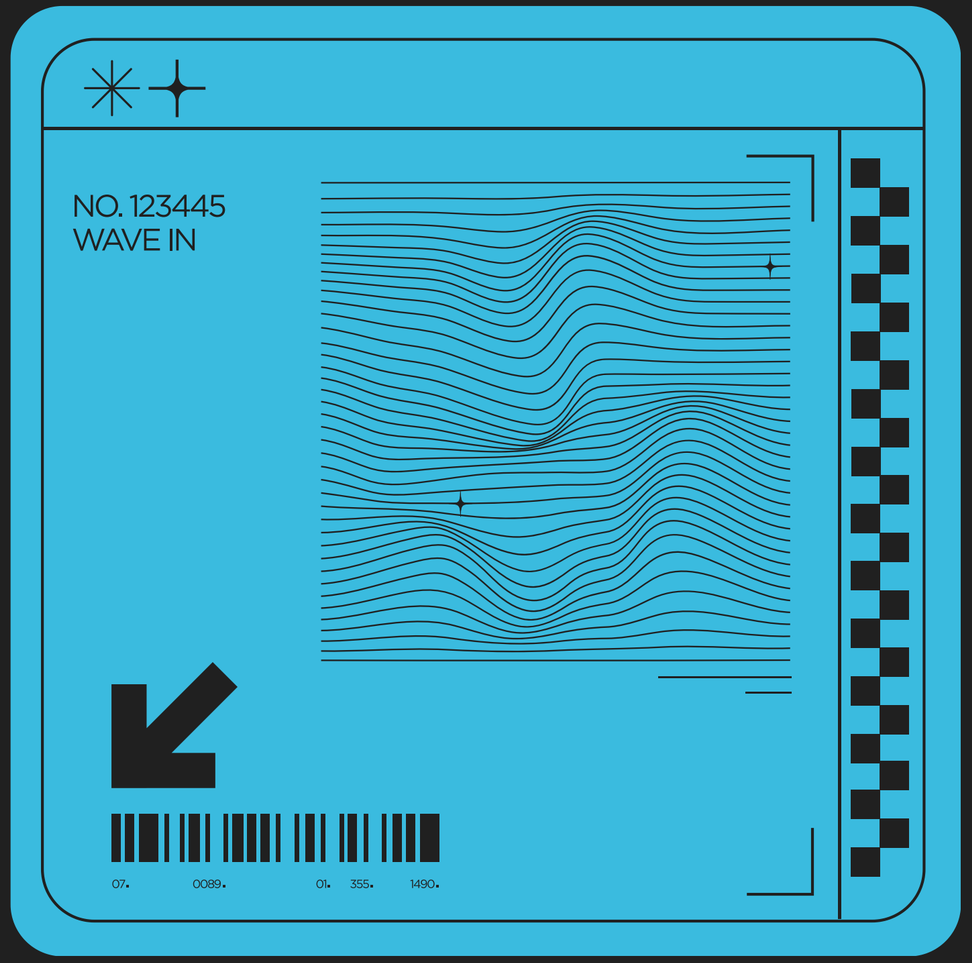
- Objective: Develop an AI-ML model to detect phishing websites using a dataset of 32+ parameters.
- Dataset: Consists of 11,000+ labeled samples of phishing and legitimate websites.
- Parameters: Include URL length, age of domain, SSL certificate, etc.
- Approach: Use machine learning algorithms (e.g., logistic regression, decision trees) for training the model.
- Feature Selection: Apply techniques like information gain and chi-square to select relevant parameters.
- Preprocessing: Handle missing values, normalize features, ensure data quality.
- Progress: Collected and cleaned dataset, split into training and testing sets.
- Results: Initial experiments will show model accuracy of 96+% on the validation set.

FLOWCHART


PROBLEM STATEMENT
Create an intelligent system using AI/ML to detect phishing domains which imitate look and feel of genuine domains.
Phishing is a malicious cyber activity where individuals or organizations are deceived into revealing sensitive information such as passwords, credit card numbers, or personal data. Typically, this deception occurs through seemingly legitimate communication, often via email, websites, or messages.
Collection Of Dataset
Collecting a dataset with over 10000 samples and 30 different features to train our ML model.
Model Training using the Collected Dataset
In the process of training our phishing detection model , we tried different ML algorithms like the random forest, decision trees , logistic regression and finally Ensemble Classifier which gave us an aggregate accuracy score of 97.4% on tuning it with accurate parameters .
We then converted it to a pkl file that could be integrated through flask in our project .
Data Preprocessing and Model Training
Ensemble Classifier and Parameter Tuning
Model Serialization Using Pickle and Integration with Flask
Testing and Validation
Extracting 30 features
1. Usinglp()
2. longUrl()
3. shortUrl()
4. symbol()
5. redirecting()
6. prefixSuffix()
7. subDomains()
8. check_https()
9. domainRegLen()
10. favicon()
11. nonStdPort()
12. httpsDomainURL()
13. requestURL()
14. anchorURL()
15. linksInScriptTags()
16. serverFormHandler()
17. infoEmail()
18. abnormalURL()
19. websiteForwarding()
20. statusBarCust()
21. disableRightClick()
22. usingPopupWindow()
23. iframeRedirection()
24. ageofDomain()
25. dnsRecording()
26. websiteTraffic()
27. pageRank()
28. googleIndex()
29. linksPointingToPage()
30. statsReport()
Creating a functioning website for checking urls
NotSoPhishy is created with HTML , CSS and JavaScript and already has been deployed for usage.
USER ENTERS THE URL TO BE CHECKED IN THE SEARCH BAR
RESULT IS DISPLAYED
CHECK ANOTHER URL
CHECKS IF IT IS A PHISHING SITE
YES
NO
Result: THIS WEBSITE IS SAFE TO USE.
Result: THIS WEBSITE IS A POTENTIAL PHISHING WEBSITE.
It also displays the valid reasons due to which the url is a phishing .
Added feature of Chrome Extension
Our project features a Chrome extension that enhances browsing safety. By simply hovering the cursor over a website URL, a popup appears, providing instant information about whether the site is potentially phishing or not, eliminating the need to click on the link for verification.
By combining real-time analysis, intuitive display, and additional information options, our Chrome extension aims to enhance user browsing safety and minimize the risk of falling victim to phishing attacks.
STEPS:
Hover Over URL:
When the user hovers the cursor over a URL, the extension triggers the real-time analysis process.
Popup Display:
A compact yet informative popup appears near the hovered URL, providing an initial assessment of the website's safety.
Color-coded Alert:
The popup is color-coded, with green indicating a safe site and red indicating a potentially phishing site.
Additional Details:
Users can click on the popup for more details, revealing the features and factors that influenced the assessment.

IDEAS/ APPROACH

- Phishing is the most common cyber threat today, exploiting user vulnerability rather than system weaknesses.
- Social engineering, relying on human error, is more potent than hacking methods.
- Our solution involves providing users with tools to make better decisions, using systems to identify complex patterns.
- An ML model trained on 11,000+ samples predicts URL reliability to detect phishing sites before user interaction.
- Our website offers a list of identified phishing URLs, global attack reports, and a search feature for users to check suspicious URLs.

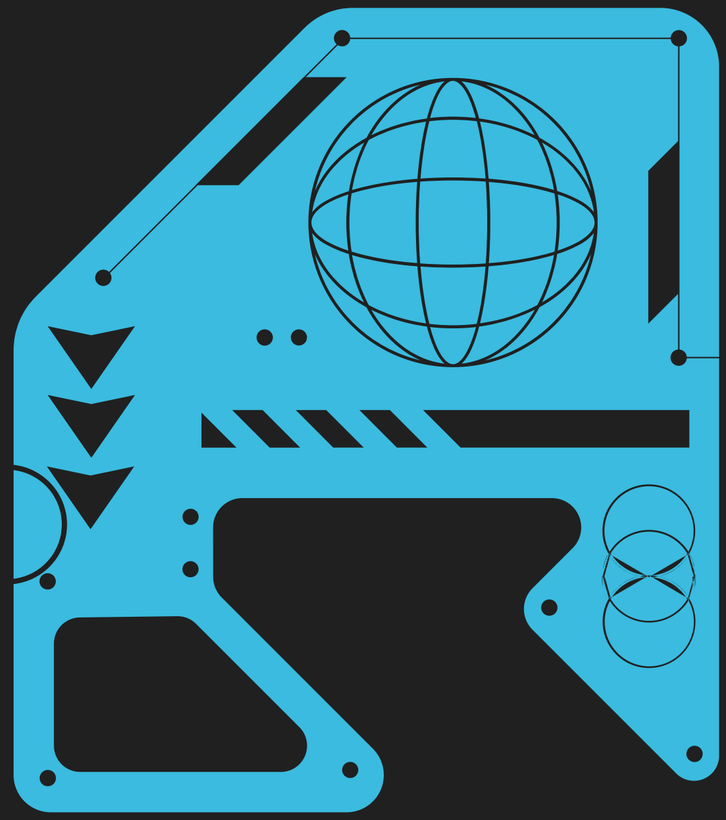


DEPENDENCIES AND NOVELTY

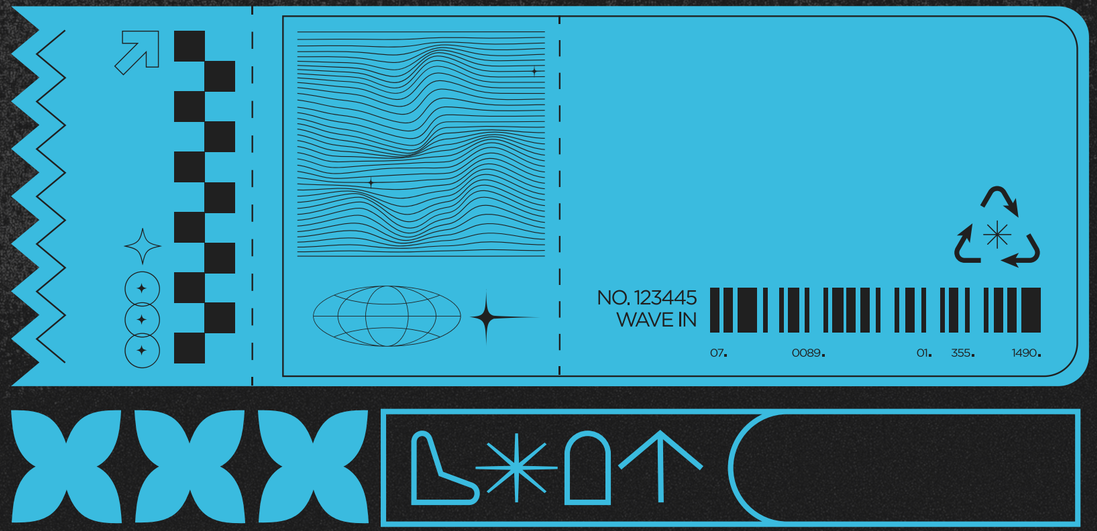
- We are focusing on including the entire functionality of this project as a google extension to increase the ease of use of the project.
- Our aim with the website is for it to check the website in realtime by checking all the parameters used by our model.
- Also, we are planning to flag and report the links identified as phishing URLs.

Dependencies:-

- A large set of data to train the AI/ML model.
- A way to collect and process data from the web.
- A way to connect to other systems and services.

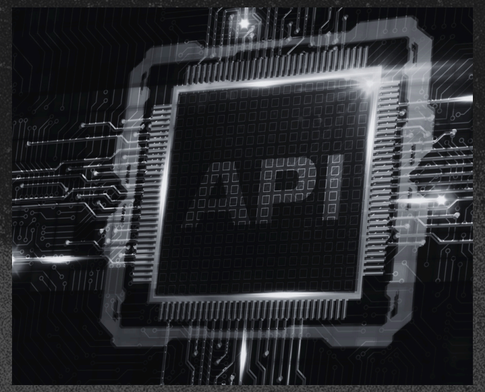


SERVICES WE WILL PROVIDE.





Real-time Phishing Website Detection



Access to our ML model through API



Google Chrome Extension



D2C FOCUSED APPLICATION

WORK PROGRESS

DATA CLEANING ↘	FEATURE EXTRACTION ↘	WEBSITE DEPLOY ↘
STATUS **COMPLETED**	STATUS **COMPLETED**	STATUS **COMPLETED**
We have completely cleaned the data, filled the missing values and preprocessed it completely to use it to train our model.	We have completely developed functions for extracting all the features that will be extracted from the website that has to be tested by our website.	We are now in process to train model and connect it to our website trough flask so users can access the prototype to check whether the website is phishy or not.

Not So Phishy

http://127.0.0.1:5000

NOT SO PHISHY

About Us | Contact Us | Get Extension

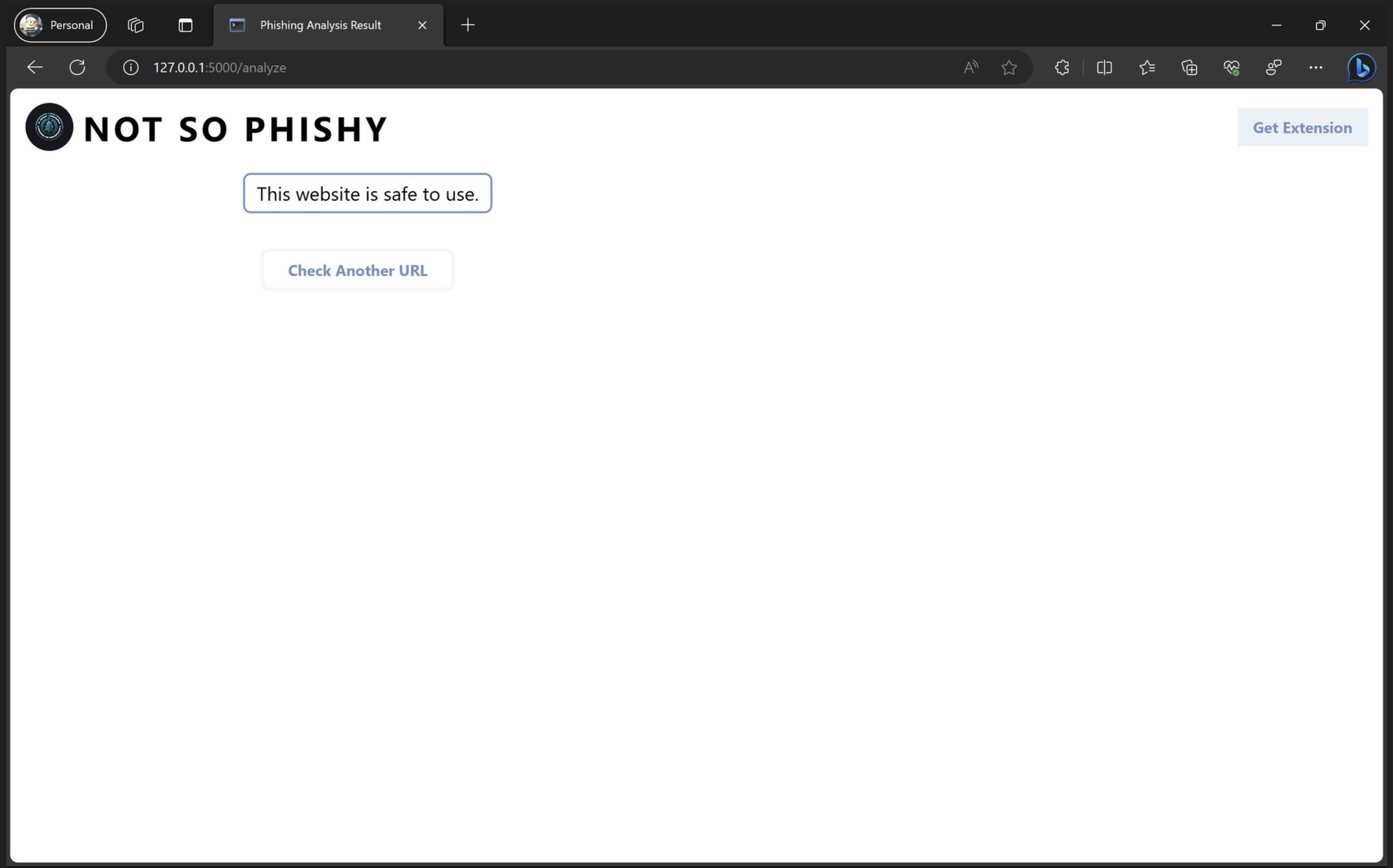
Search for information of phishing website connected to the public Internet.

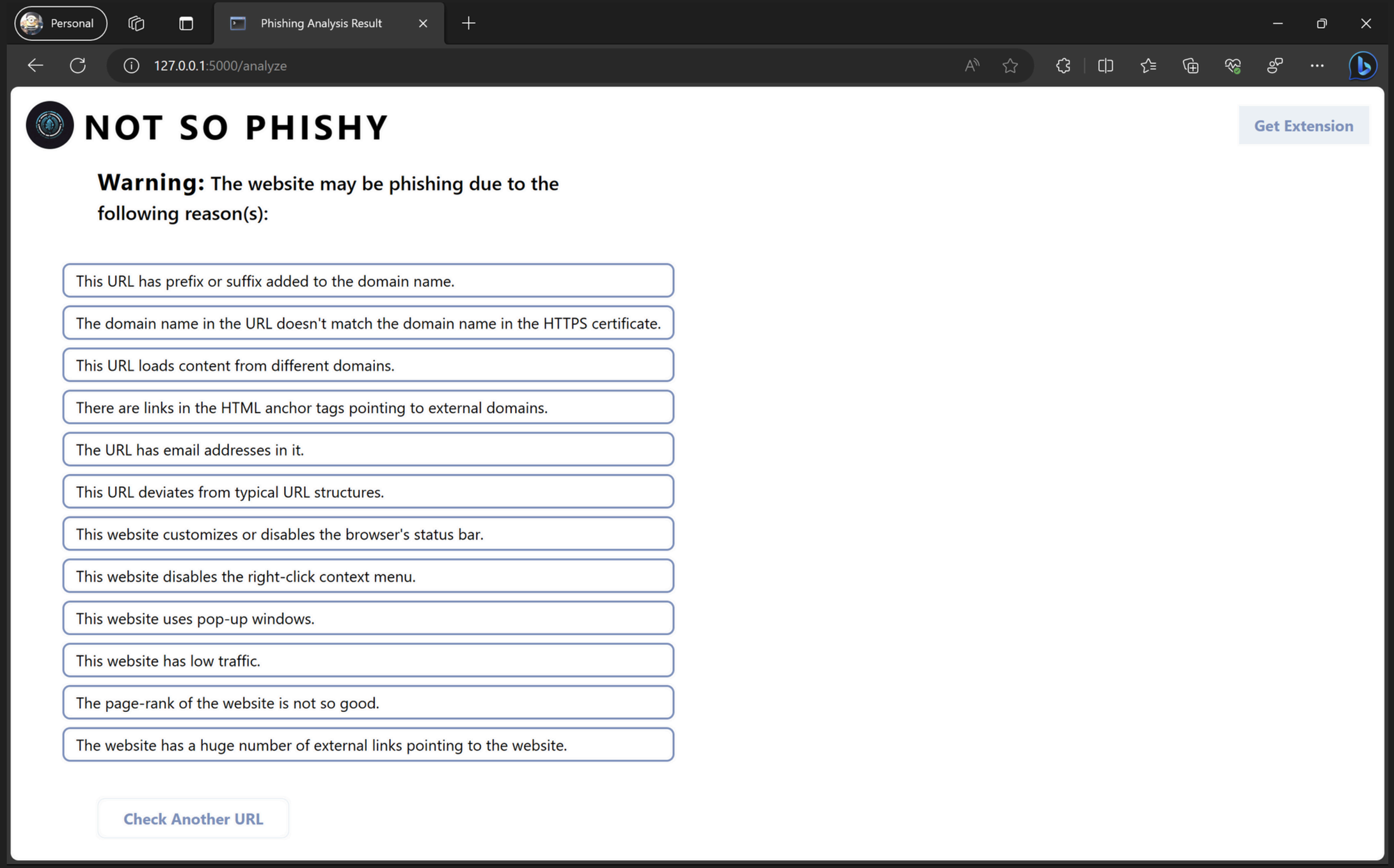
Enter The Website URL





127.0.0.1:5000/phishingsUpdates


Personal
Phishing Analysis Result
127.0.0.1:5000/analyze
NOT SO PHISHY
Get Extension
This website is safe to use.
Check Another URL


Personal
Phishing Analysis Result
127.0.0.1:5000/analyze
NOT SO PHISHY
Get Extension
Warning: The website may be phishing due to the following reason(s):
This URL has prefix or suffix added to the domain name.
The domain name in the URL doesn't match the domain name in the HTTPS certificate.
This URL loads content from different domains.
There are links in the HTML anchor tags pointing to external domains.
The URL has email addresses in it.
This URL deviates from typical URL structures.
This website customizes or disables the browser's status bar.
This website disables the right-click context menu.
This website uses pop-up windows.
This website has low traffic.
The page-rank of the website is not so good.
The website has a huge number of external links pointing to the website.
Check Another URL

THANK YOU